

L3 - Transformations of $\sqrt{x}$ - Lesson

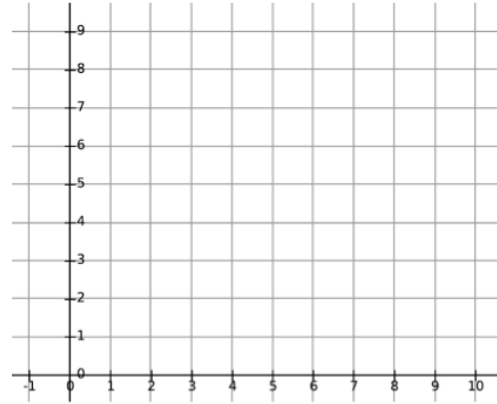
MCR3U

Base Function:

Key Points:

x	y

Graph of Base Function:



Example 1: Using the parent function $f(x) = \sqrt{x}$, describe the transformations and write the equation of the transformed function $g(x)$.

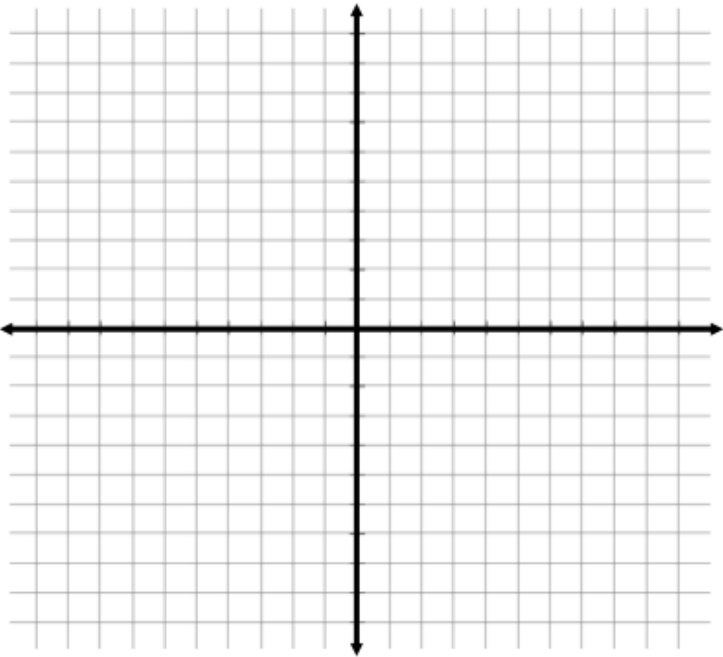
$$g(x) = -2f\left[-\frac{1}{3}(x + 6)\right] - 5$$

Example 2: for each of the following functions...

- make a table of values for the parent function
- graph the parent function $f(x) = \sqrt{x}$
- describe the transformations
- make a table of values of image points
- graph the transformed function and write its equation

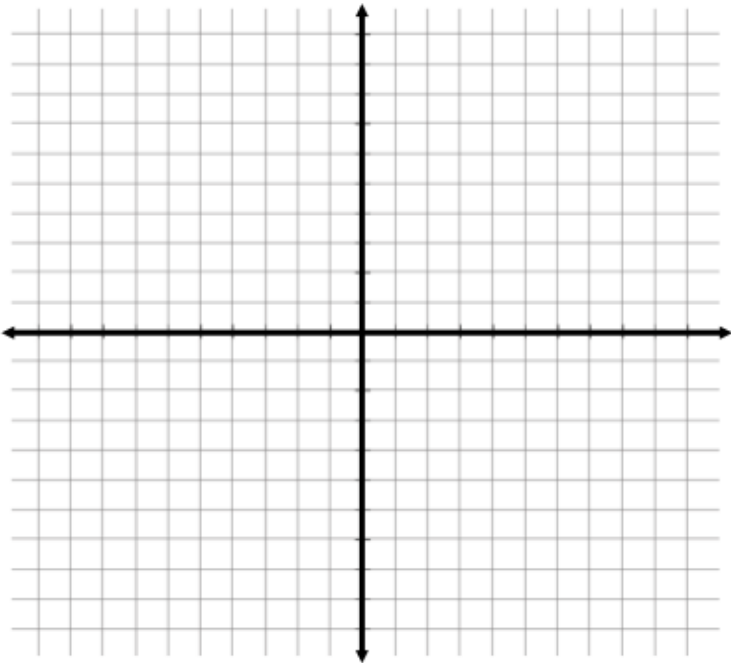
a) $g(x) = \frac{1}{2}f(x) + 1$

<i>x</i>	<i>y</i>

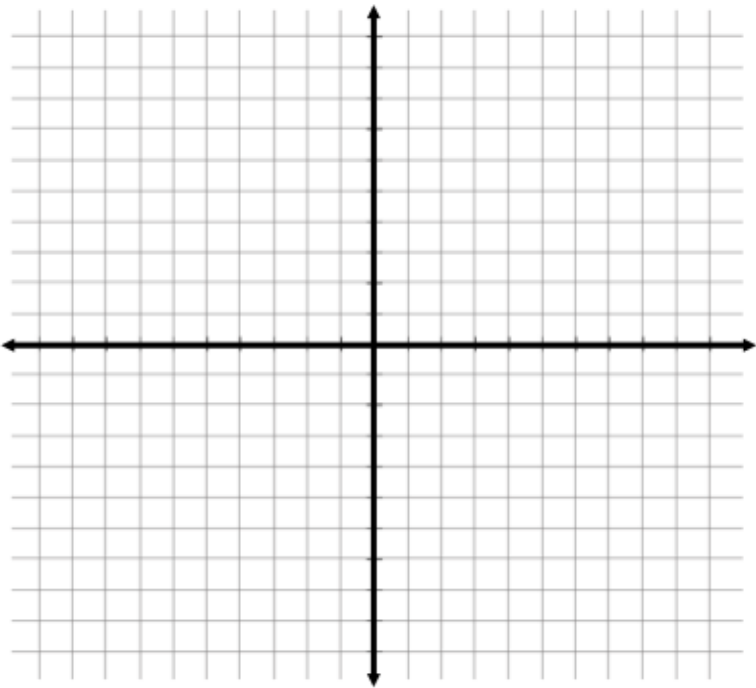


b) $g(x) = -f[2(x - 3)]$

<i>x</i>	<i>y</i>

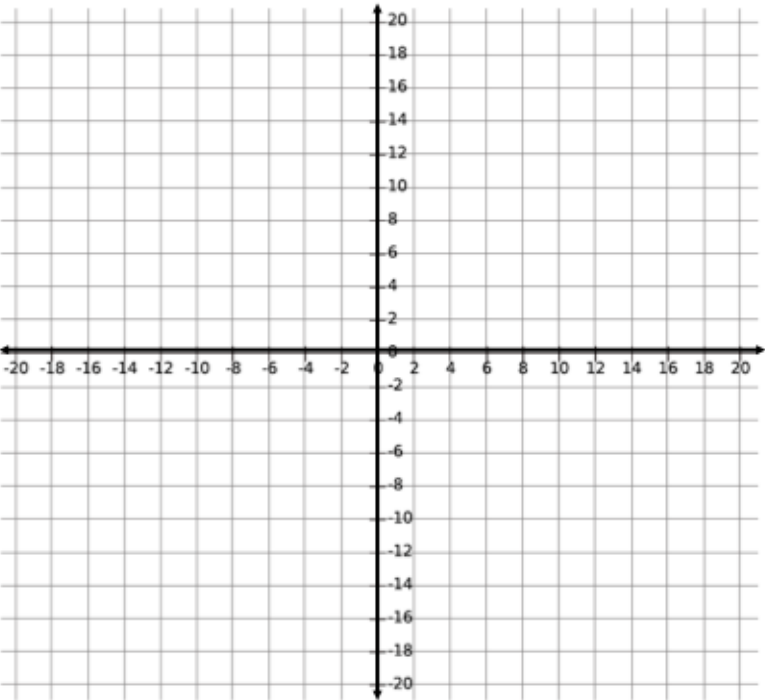


c) $g(x) = -2f(x + 3) - 1$



<i>x</i>	<i>y</i>

d) $g(x) = 3f\left(-\frac{1}{2}x + 2\right) + 1$



<i>x</i>	<i>y</i>